

LAKSHITHA MATHANGADEERA

Senior Software Engineer

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SKILLS SUMMARY

- **Backend Development & Frameworks**

- PHP MVC Frameworks: Magento 2, Laravel
- Node.js Frameworks: NestJS (REST APIs, modular architecture, authentication, RBAC)
- LEMP Stack (Linux, Nginx, MySQL, PHP): Scalable server configurations, high-performance tuning
- Magento 2 Development: Custom module development, checkout customization, order management, product attributes, and catalog enhancements
- Magento 2 Design Patterns: Factory, Repository, Dependency Injection (DI), Service Contracts, Observers, Plugins, Event-Driven Architecture
- Performance Optimization: Caching (Redis, Varnish), query optimization, index tuning
- API Development: RESTful APIs, GraphQL, Swagger/OpenAPI Documentation
- Microservices & SOA: Service-Oriented Architecture (SOA) for eCommerce systems
- Python (Flask, Django – AI-based services and admin dashboards)
- RAG Systems: Vector search (pgvector), embedding pipelines, document ingestion

- **Frontend Development & Technologies**

- Magento 2 UI Components: Knockout.js, RequireJS, PHTML Templates
- Progressive Web Apps (PWA): Magento PWA Studio, React, Redux
- Next.js (App Router), React, Tailwind CSS, ShadCN UI
- GraphQL & AJAX: GraphQL APIs for data fetching, AJAX for real-time frontend updates
- Frameworks and Headless CMS: Nextjs, Contentful, Strapi
- SSR & Performance Optimization in Next.js applications
- State Management & API integration (REST / GraphQL)

- **Cloud Hosting & DevOps**

- Cloud Services: AWS (EC2, RDS, S3, CloudFront, Lambda, Auto Scaling, Load Balancers), Alibaba Cloud, cPanel, Vercel, Netlify
- Process Management: PM2, Supervisor
- CI/CD Pipelines: Jenkins, GitHub Actions, Bitbucket Pipelines
- Containerization & Orchestration: Docker, Kubernetes
- Version Control: Git, GitHub, Bitbucket
- Content Delivery & Security: Cloudflare, AWS CloudFront for CDN and DDoS protection

- **Architecture & System Design**
 - Designed scalable **SOA and microservice-based architectures**
 - Experience with **high-concurrency systems and distributed services**
 - Built **multi-tenant SaaS platforms with strict data isolation**
 - Implemented **event-driven and queue-based processing systems**
 - **Security & Performance**
 - Multi-tenant architecture design and isolation strategies
 - API rate limiting, request validation, and secure token handling
 - CDN + caching strategies (Cloudflare + CloudFront + Varnish)
 - Advanced Security Measures: Data encryption, CSRF/XSS protection, Content Security Policy (CSP)
 - SEO Best Practices: Structured data, meta tags, schema markup
 - Performance Monitoring: New Relic, Blackfire.io, AWS CloudWatch
 - **Testing & Debugging**
 - Unit Testing & TDD: PHPUnit, PestPHP
 - Functional & Integration Testing: Selenium, Cypress, Playwright
 - Performance Testing: Grafana K6, JMeter, Locust
 - **Database Management**
 - Relational Databases: MySQL, PostgreSQL
 - NoSQL Databases: MongoDB, DynamoDB
 - Database Optimization: Indexing, Query Optimization, Read Replicas
 - Vector Databases: PostgreSQL with pgvector (AI search use cases)
 - Multi-tenant database design and optimization
 - High-performance query tuning for large-scale datasets
 - **Soft Skills**
 - Leadership & Team Collaboration
 - Problem Solving & Analytical Thinking
 - Effective Communication & Mentoring
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WORK EXPERIENCE

Senior Software Engineer – Azbow PVT Limited

April 2025 – Present

- Designed and implemented a **multi-tenant SaaS platform** for AI-powered customer support and data querying.
- Built a **RAG-based chatbot system** using document ingestion, embeddings, and vector search for accurate responses.

- Integrated **WhatsApp Cloud API** to enable automated customer interactions with AI-driven responses.
- Developed backend services using NestJS with modular architecture, ensuring scalability and maintainability.
- Built a Next.js frontend with Tailwind CSS for admin dashboards and user-facing applications.
- Designed **queue-based background processing** for file ingestion, data syncing, and async jobs.
- Managed **cloud infrastructure on AWS EC2**, including deployment, scaling, and monitoring.
- Optimized system performance using **Redis caching, CDN (Cloudflare), and query tuning**.
- Built a Next.js frontend with Tailwind CSS for admin dashboards and user-facing applications.
- Developed internal tools for **report generation, PDF processing, and automation workflows**.

Senior Software Engineer – Efutures PVT Limited

September 2024 – April 2025

- Led the Magento 2 development team for Airport Home Appliances (airportappliance.com), a high-traffic US-based eCommerce store.
- Developed custom modules, checkout enhancements, and performance optimizations.
- Implemented Redis caching, Elasticsearch, and Varnish for faster load times and scalability.
- Managed AWS-based hosting and server optimizations for better uptime and reliability.
- Integrated custom ERP solutions for real-time inventory, sales, and customer data sync.
- Mentored junior developers, ensuring best coding practices and high-quality deliverables.

Associate Software Manager- NeoSolax PVT Limited

Oct 2020 - Sep 2024

- Built and maintained multiple high-traffic Magento 2 eCommerce websites, including Buyabans, Nomin, Bonz, J20, Chihuly, Arienti, Stripes and Checks, Kidooz, Biopak, and Vereina-Shop.
- Performance Optimization: Improved MySQL queries, Redis caching, and Varnish configurations to enhance site speed.
- Custom ERP & API Integrations: Implemented real-time stock, sales, and customer synchronization with external ERP systems.
- Payment Gateway Customization: Developed custom payment methods supporting bank-specific discounts and promotions.
- Service-Oriented Architecture (SOA) Setup: Designed scalable hosting solutions using AWS, Alibaba Cloud, and cPanel-based infrastructures.

PROJECTS

- Full-Stack Property / SaaS Application (NestJS + Next.js)
Developed backend using NestJS and MongoDB for scalable API services.
Built responsive frontend using Next.js and Tailwind CSS.
Implemented authentication, CRUD operations, and deployment pipelines.
- AI-Powered RAG SaaS Platform
Built a multi-tenant SaaS platform to process documents and provide AI-based answers.
Implemented document ingestion pipeline with chunking and embedding storage (pgvector).
Developed chat API to return contextual answers with citations.
Integrated WhatsApp API for real-time AI-based customer communication.
- Magento AI Chatbot Integration
Integrated chatbot into Magento admin to query business data (e.g., sales, orders).
Connected LLM with Magento database through a custom MCP server.
Enabled real-time business insights through natural language queries.
- Airport Home Appliances (2024 - Present) | Magento 2, AWS (2024 -2025)
Led Magento 2 development and performance optimizations for a high-volume US-based store.
Implemented custom checkout features, ERP integration, and caching improvements.
- Build and Maintenance of Various Magento 2 Projects (2020 - 2024)
Integrated Magento 2 with multiple ERP systems for real-time inventory and sales data synchronization, developed custom payment solutions with dynamic bank discounts and real-time promotions, and contributed to high-traffic eCommerce platforms including Nomin (PWA Project), Buyabans, Bonz, Kidooz, J20 Sports, Chihuly, RHB, FHM, Arienti, Biopak, Stripes and Checks, and KDG Cars.

EDUCATION

- BSc (Hons) in Information Technology
Sri Lanka Institute of Information Technology (SLIIT), Malabe 2016 - 2020

REFERENCES

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